

RAHUL PAUL

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EDUCATION

MS Ramaiah Institute of Technology

B.E. Computer Science and Engineering (Cyber Security), CGPA 9.52/10.0

Bengaluru, Karnataka

2023 – 2027

Bhavan's Gangabux Kanoria Vidyamandir

Class XII (AISSCE), 96.33%

Kolkata, West Bengal

2021 – 2022

EXPERIENCE

Samsung R&D Institute India

Project Intern, Samsung PRISM

Bengaluru, Karnataka

Dec 2025 – Jul 2026

- Took EmoCapNet from research prototype to an on-device build by cutting its size and latency. Distilled a ViT + GPT-2 captioner into a TinyCLIP-ViT-8M + DistilGPT-2 student and applied quantization-aware training for INT8, reaching **6.8× smaller (146 MB) and ~450 ms per caption on CPU**, with from-scratch KV-cache decoding.
- Made captions emotion-controllable through a continuous valence-arousal-dominance (VAD) vector with FiLM conditioning instead of discrete emotion labels, **halving directional error** against a discrete-label baseline.

PROJECTS

Smart Course Generator · [GitHub](#) · [Live Demo](#)

May 2026 – Jun 2026

- Engineered a custom multi-provider AI router (Gemini → Groq → OpenRouter) with per-provider circuit breakers, retry with backoff, and timeouts, so an upstream outage or rate limit **degrades the response instead of failing it**.
- Cut perceived latency by **streaming courses lesson-by-lesson over SSE**, so users see first output in seconds instead of waiting on a multi-minute generation. Secured it with rotating refresh tokens and per-route rate limiting.
- Built the quality gate alongside the feature: **unit and integration tests, Playwright end-to-end suites, and autocannon load tests in CI**, plus an LLM-as-judge eval harness that fails the build on output-quality regressions.

Tech: React, TypeScript, Node.js, Express, MongoDB, SSE, Zod, Jest, Playwright, autocannon, GitHub Actions

Three-Way Financial Reconciliation Engine · [GitHub](#) · [Live App](#) · [Video](#)

Aug 2026 – Sep 2026

- Built a data-processing engine that reconciles a merchant order ledger, a payment-gateway report, and a bank statement through tiered deterministic matchers, resolving **90.8% of 141 entities at 100% accuracy** across 14 defect classes. Every unresolved record lands in a categorised exception report with its reason.
- Implemented the $O(n^3)$ Hungarian algorithm from scratch to resolve contested bank-to-settlement matches jointly rather than greedily, behind a verification gate encoding paise-exact money, MDR, and T+1 settlement rules.
- Sustains **~233K entities/sec**, flat from 141 to 5,022 entities, and is guarded by **111 tests whose adequacy was verified by mutation testing** plus a CI pipeline that asserts the measured accuracy, so a regression fails the build.

Tech: Python, FastAPI, Uvicorn, pytest, mutation testing, GitHub Actions, Render

DNSentinel, DNS Threat Detection Platform · [GitHub](#) · [Live Demo](#)

Apr 2026 – Jun 2026

- Built a real-time detection platform: a FastAPI service over SQLAlchemy with 23 REST endpoints and **SSE live streaming** to a React dashboard. Availability is handled explicitly: a liveness probe returns 503 when the database degrades, and containment rules auto-expire after 24h so a false positive cannot wedge the network.
- Traced 407 false positives to the default 0.5 decision threshold, not the features, and recalibrated against an explicit false-positive budget: **407 → 21 at unchanged 100% recall** on 12,000 domains across 4 DGA families.

Tech: Python, FastAPI, SQLAlchemy, SQLite, React, scikit-learn, Docker Compose, pytest, CodeQL

TECHNICAL SKILLS

- **Languages:** C/C++, Python, JavaScript, TypeScript, SQL, Bash
- **CS Foundations:** DSA, Object-Oriented Design, Operating Systems, Computer Networks, DBMS
- **Backend & Reliability:** FastAPI, Node.js/Express, REST, SSE, circuit breakers, retries, rate limiting
- **Testing:** unit/integration (pytest, Jest, Vitest), end-to-end (Playwright), load (autocannon), mutation
- **Data & Tooling:** MongoDB, SQLite, SQLAlchemy, schema design & indexing, Docker, Git, GitHub Actions

PUBLICATIONS

Applications of Quantum Computing for Internet of Things · Springer Nature · [Publication](#)

2025

- Co-authored a 43-page book chapter (pp. 79–121) in *Quantum Computing, Sensing and Communications for IoT*, surveying 70+ papers across quantum ML, post-quantum cryptography, and sensing.

ACHIEVEMENTS & CERTIFICATIONS

- **LeetCode:** Knight, max contest rating **1858** (top 6.13% globally, 22 rated contests) · [Profile](#)
- **Campus Ambassador**, Kshitij, IIT Kharagpur · [Certificate](#)
- **Certifications:** [IBM Threat Intelligence & Hunting](#), [IBM Security Operations in Practice](#), [Infosys Springboard DSA](#)